

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

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1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #75. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #75 - [P0] Biomarker Prevalence Not Treated as Uncertain Parameter
- **Baseline:** Commit 2b04ea7 (2025-12-10 15:23:08)
- **Fixed:** Commit c15aa47 (2025-12-10 17:13:53)

2 Metadata Comparison

Table 1: Metadata Comparison

Item	Baseline	Fixed
Git Commit	2b04ea7	c15aa47
Timestamp	2025-12-10 15:23:08	2025-12-10 17:13:53
R Version	R version 4.5.1 (2025-06-13 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	75	75

2.1 Global Parameters

Table 2: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 3: Base Case Incremental Cost Comparison

Strategy	Baseline (€)	Fixed (€)	Difference (€)
control	0.00	0.00	0.00
crp	33,930.80	33,930.80	0.00
tlr	64,024.26	64,024.26	0.00
tmb_braf	41,540.41	41,540.41	0.00

Table 4: Base Case Incremental Effect Comparison

Strategy	Baseline (QALYs)	Fixed (QALYs)	Difference (QALYs)
control	0.0000	0.0000	0.0000
crp	-0.0599	-0.0599	0.0000
tlr	-0.1406	-0.1406	0.0000
tmb_braf	0.0252	0.0252	0.0000

Table 5: Base Case ICER Comparison

Strategy	Baseline (€/QALY)	Fixed (€/QALY)	Difference (€/QALY)
control	—	—	—
crp	—	—	—
tlr	—	—	—
tmb_braf	1,645,887	1,645,887	0

3.1 Efficiency Frontier Comparison

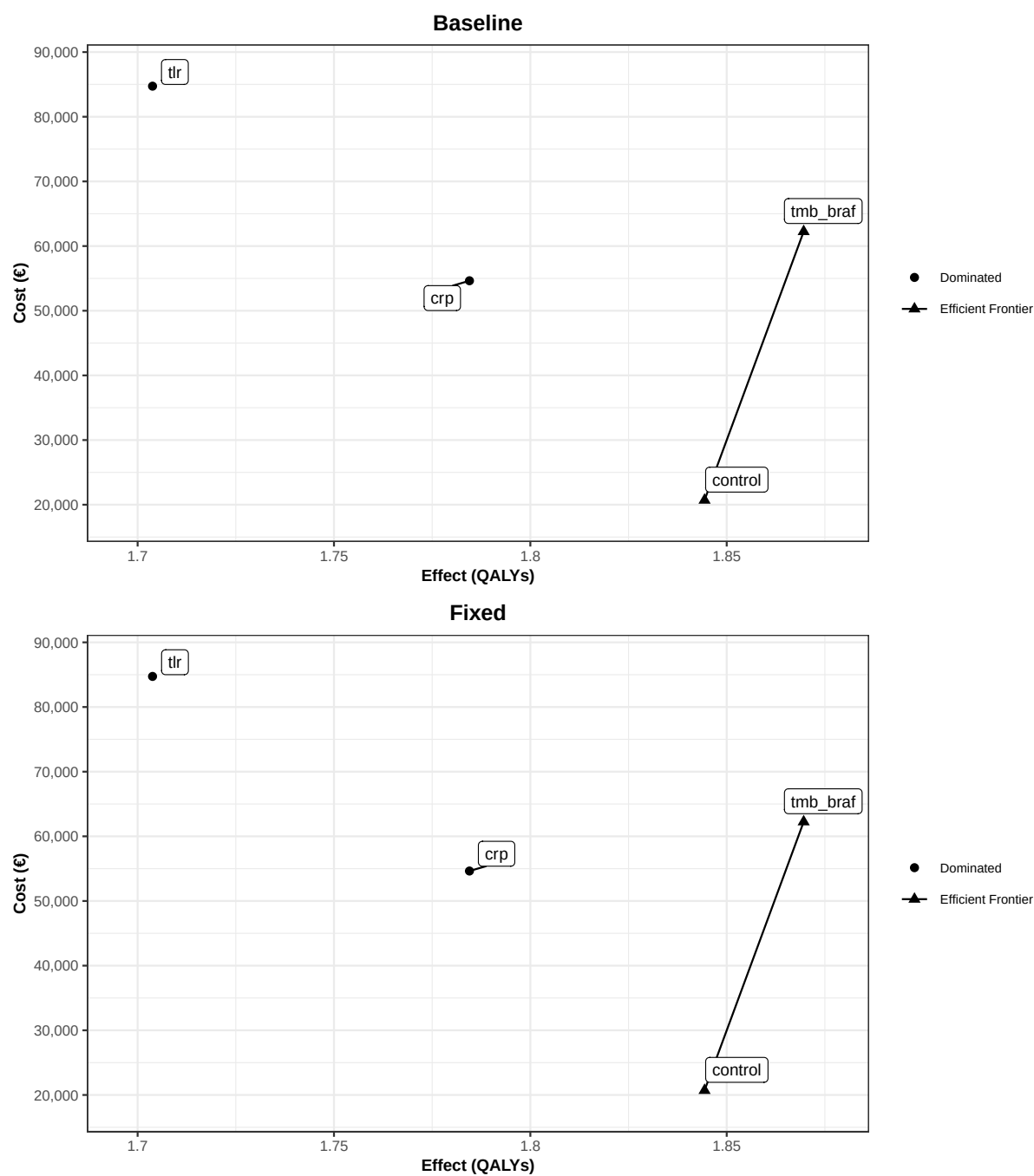


Figure 1: Cost-effectiveness efficiency frontier comparison using dampack. Baseline (top) and fixed (bottom) strategies are shown with the efficiency frontier connecting non-dominated strategies. Dominated strategies appear off the frontier line.

3.2 Net Monetary Benefit Comparison

Optimal Strategy: Unchanged (control)

Table 6: Net Monetary Benefit Comparison at WTP = €51,000

	Strategy	Baseline (€)	Fixed (€)	Difference (€)
1	control	73,356.74	73,356.74	0.00
2	crp	36,372.76	36,372.76	0.00
4	tmb_braf	33,103.52	33,103.52	0.00
3	tlr	2,163.83	2,163.83	0.00

4 PSA Results Comparison

Table 7: PSA Cost Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (€)
control	€20,929.38 (€15,783.79; € 27,127.22)	€20,929.38 (€15,783.79; € 27,127.22)	0.00
crp	€54,765.25 (€41,298.43; € 70,620.25)	€54,852.98 (€39,451.38; € 74,339.21)	87.73
tlr	€87,131.61 (€62,251.12; €116,305.00)	€87,302.96 (€57,700.43; €124,136.37)	171.36
tmb_braf	€65,672.39 (€48,646.95; € 85,670.59)	€65,618.30 (€45,504.96; € 91,585.38)	-54.09

Table 8: PSA Effects Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (QALYs)
control	1.6974 (1.1101; 2.3477)	1.6974 (1.1101; 2.3477)	0.0000
crp	1.8749 (1.2513; 2.4153)	1.8762 (1.2516; 2.4180)	0.0013
tlr	1.8052 (1.1909; 2.3626)	1.8078 (1.1626; 2.4313)	0.0026
tmb_braf	1.9755 (1.2785; 2.6461)	1.9754 (1.2768; 2.6651)	-0.0001

Table 9: PSA ICER Results (n = 5,000 simulations)

Strategy	Baseline Mean ICER	Fixed Mean ICER
control	—	—
crp	€190,672/QALY	€189,725/QALY
tlr	€614,388/QALY	€601,344/QALY
tmb_braf	€160,903/QALY	€160,742/QALY

Note: ICERs calculated from PSA mean costs and effects as point estimates.

4.1 PSA Visual Analysis

4.1.1 PSA Scatter Plots (Side-by-Side)



Figure 2: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

4.1.2 PSA Scatter Plots (Overlaid)

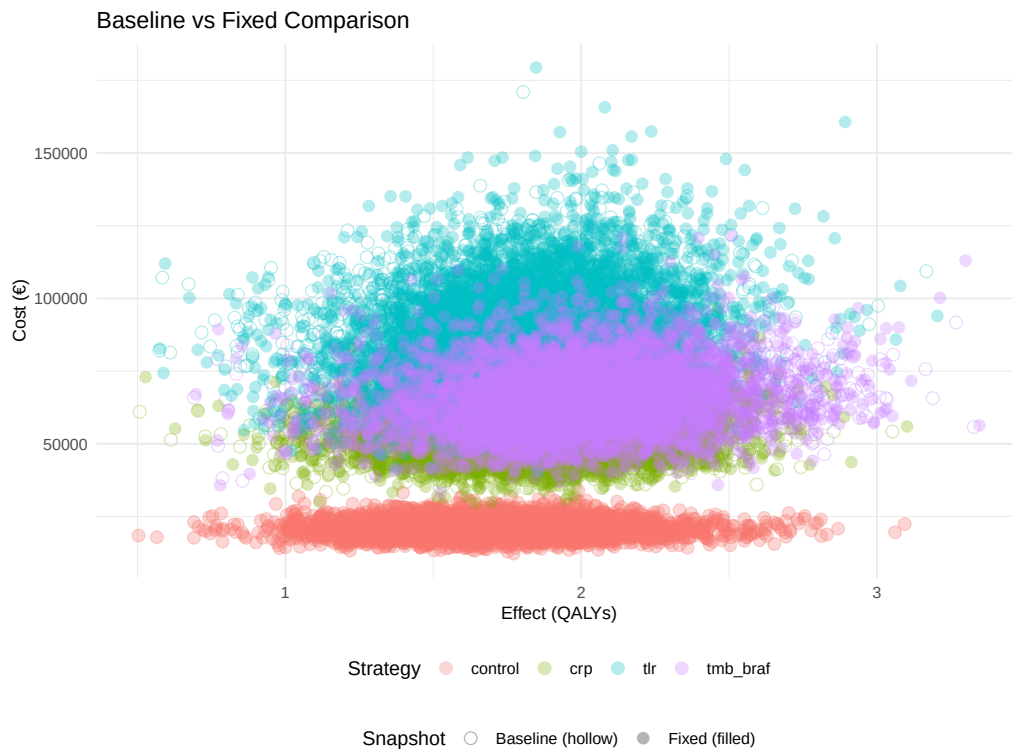


Figure 3: Overlaid cost-effectiveness plane comparing baseline and fixed PSA simulations. This plot overlays all PSA simulation results from both snapshots on a single plane, where each point represents one PSA iteration for a given strategy. Hollow circles indicate baseline simulations, while filled circles indicate fixed simulations. Colors distinguish strategies, while shapes distinguish snapshots. The overlay allows direct visual comparison of how the probabilistic distributions changed after the bug fix: if the fix was purely methodological with no impact on central estimates, baseline and fixed points would overlap completely, whereas systematic shifts in the point clouds indicate changes in underlying model behavior.

5 Comparison Across Recent Issues

5.1 Available Snapshot Pairs

Table 10: All Available Snapshot Pairs

Issue		Title	Baseline Commit	Fixed Commit
7	75	[P0] Biomarker Prevalence Not Treated as Uncertain Parameter	2b04ea7	c15aa47
11	90	Sensitivity analysis: independent biomarker-specific survival models	9e1b590	7d68bca
8	76	[P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)	64cf543	64cf543
10	87	[P0] Fix PSA methodology: Replace single-patient sampling with subgroup population averaging	42d4acb	0a15902
9	86	[P1] Mltpl documentation issues	ee96c31	f3e5604
6	70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	998dacd	605ff90
5	68	[P4] Include Weibull model for reference	4665f93	998dacd
1	42	[P0] Critical Bug: Incorrect death transition calculation in calcu- late_outcomes.R	607b539	cadf81a
4	67	[P4] Move snapshot folder	f3674e8	d5d1356
2	62	[P4] Move cached file	d046c7c	f3674e8
3	64	[P3] impact of bug fixes	8681d1e	8681d1e

5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 11: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Opt. (BL)	Opt. (Fix)	Chg?	Max Cost Δ (€)	Max Effect Δ (QALYs)	Max ICER Δ (€/QALY)
75	[P0] Biomarker Prevalence Not Treated as Uncertain Parameter	control	control	No	0.00	0.0000	0
90	Sensitivity analysis: independent biomarker-specific survival models	control	control	No	0.00	0.0000	0
76	[P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)	control	control	No	0.00	0.0000	0
87	[P0] Fix PSA methodology: Replace single-patient sampling with subgroup population averaging	control	control	No	0.00	0.0000	0
86	[P1] Mltpl documentation issues	control	control	No	0.00	0.0000	0

6 Summary

The comparison for issue #75 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** €0/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

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